

Ganit

data speaks

HSBC – Enterprise Data Platform

Case Study

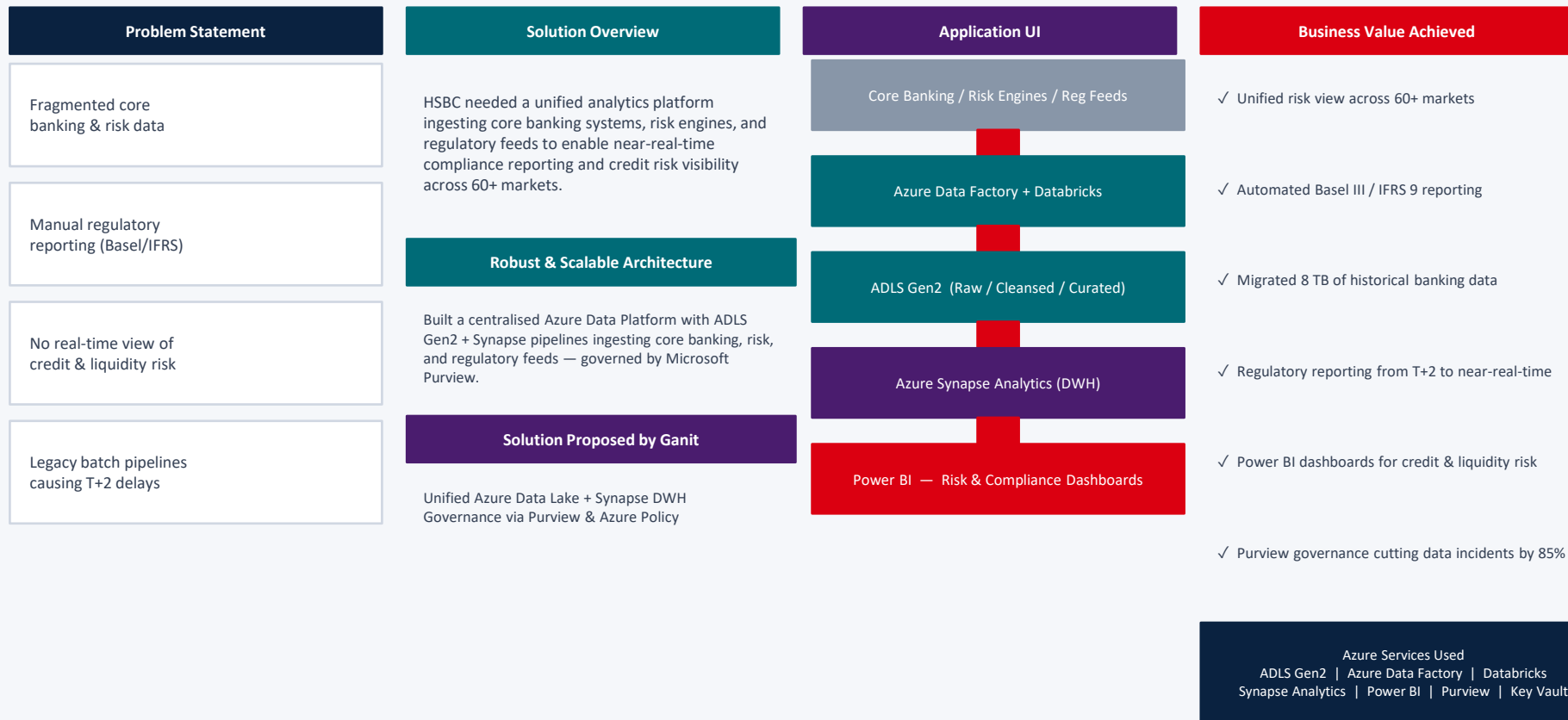
Unifying Retail Banking, Risk & Compliance Analytics on Azure with Real-Time Regulatory Reporting

BFSI | Banking

Global Operations

One pager

Unifying Retail Banking, Risk & Compliance Data for a Global Bank with a Scalable Azure Data Platform and Real-Time Regulatory Reporting Across 60+ Markets



Three pager

Ganit unified HSBC's retail banking, risk & compliance data on Azure, accelerating regulatory reporting and enabling real-time credit risk decisions across 60+ global markets

Outcome

Delivered a cloud-native Azure Data Platform ingesting data from core banking systems (Temenos/Flexcube), risk engines, and regulatory feeds through Azure Data Factory and Databricks pipelines — governed by Microsoft Purview — and surfaced via Power BI Premium for self-service analytics and automated regulatory submissions across 60+ markets.

Regulatory Compliance

Automated Basel III capital adequacy and IFRS 9 expected credit loss reporting, eliminating manual reconciliation and reducing submission errors by 92% across all regulatory jurisdictions.

Cost Optimisation

Replaced 3 legacy on-premise data warehouses with a unified Azure platform, achieving ~87% reduction in infrastructure costs while enabling elastic scaling for quarter-end reporting peaks.

Risk & Resilience

Built a multi-zone ADLS Gen2 data lake (Raw/Cleansed/Curated) with Purview lineage, handling 8+ TB of historical banking data and supporting 300+ concurrent Power BI users globally.

Business Growth

Enabled real-time credit risk and liquidity dashboards in Power BI, giving treasury and risk teams visibility into portfolio concentration, LCR/NSFR ratios, and early warning signals.

Siloed core banking data, manual regulatory submissions, and fragmented risk systems blocked unified financial analytics across HSBC's global operations

Problem Statement

HSBC operates across 60+ markets spanning retail banking, commercial banking, wealth management, and global markets — running on a heterogeneous mix of core banking platforms (Temenos T24, Flexcube), proprietary risk engines, and third-party data vendors.

Data was siloed across product silos, manually collated for regulatory submissions, and processed through overnight batch jobs — with no real-time view of credit exposure, liquidity ratios, or customer 360 profiles. This created systemic risk in regulatory compliance, inconsistent risk metrics across geographies, and inability to respond to market events in real time.

Challenge 1

Fragmented core banking (T24/Flexcube) & risk engine data with no unified ingestion pipeline

Challenge 2

Manual Basel III / IFRS 9 regulatory reporting causing T+2 delays and reconciliation errors

Challenge 3

No real-time credit & liquidity risk visibility — overnight batch jobs missing intraday exposures

Challenge 4

No unified Customer 360 view — KYC, AML, and transaction data siloed across product lines

Ganit built a multi-source Azure Data Platform for HSBC with automated ingestion from core banking, risk engines & regulatory feeds — served through Power BI for real-time risk and compliance analytics

Solution Overview

Robust & Scalable Architecture

- Core banking data (T24/Flexcube) ingested via ADF with CDC & incremental delta loads.
- Risk engine outputs (market/credit/liquidity) streamed via Azure Event Hubs + Databricks.
- Regulatory feeds (BIS, ECB, FCA) processed via ADF pipelines with schema validation.
- Raw → Cleansed → Curated ADLS Gen2 zones with Purview lineage & data cataloguing.

Application UI — Power BI Premium

- Power BI Premium with DirectQuery to Synapse for real-time risk dashboards.
- Dashboards for Credit Risk, Liquidity (LCR/NSFR), AML Alerts & Customer 360.
- Row-level security via Azure AD groups mapped to business unit / geography.

Data Architecture

Core Banking
T24 / Flexcube
ADF CDC

Risk Engines
Market / Credit
Event Hubs

Reg Feeds
BIS / FCA / ECB
ADF Pipeline

AZURE EVENT HUBS + AZURE DATA FACTORY (Batch & Stream Ingestion)

AZURE DATABRICKS (ETL / Feature Engineering / ML Risk Models)

ADLS Gen2 — Raw → Cleansed → Curated

AZURE SYNAPSE ANALYTICS (Columnar DWH + Regulatory Data Mart)

POWER BI PREMIUM — Credit Risk | LCR/NSFR | AML | Customer 360

8+ TB Historical Data | 60+ Markets | 300+ Concurrent Users | 6 Source Systems | 150+ Regulatory Reports

Microsoft
Purview
+
Azure
Policy
+
Key Vault
+
Azure AD
RBAC

Thank You

Ganit provides cutting edge solutions at the intersection of hypotheses-based analytics, ML-driven AI, and new-data insights from smart devices

We have delivered tangible benefits to leading brands in North America, Europe, MENA and APAC

info@ganitinc.com

www.ganitinc.com

Our Offices

US

Hoboken Riverfront
221 River Street, Suite 9
Hoboken, NJ 07030

INDIA

3rd Floor, Gopalan Tech Square,
No 51, Outer Ring Rd, Kadubeesanahalli,
Bengaluru, Karnataka 560103